

SHRUTI JAGDALE

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EXPERIENCE

Machine Learning Intern (Agentic Systems), Zeblok Computational

May 2025 – Present

- Implemented a scalable multimodal (text, image) retrieval system for enterprise runbook documents using foundation models (LLaMA-70B), ChromaDB, MiniLM, and CLIP; integrated Model Context Protocol tool calling for function calling, RAG with vector search, and PDF parsing (PyMuPDF).
- Automated knowledge distillation by vectorizing text and image encoders in the orchestration layer, improving cosine similarity scores by 30% for downstream foundation model fine-tuning.
- Engineered ML tooling and infrastructure (SQLite-based chat memory) with CRUD logic, enabling robust CoT evaluation, privacy-preserving query handling, and reducing retrieval time by 20%.
- Containerized and deployed microservices on Zeblok's cloud platform using Docker and Kubernetes, integrated into CI/CD pipelines with FastAPI endpoints, ensuring scalable and smoother workflows.

NLP Intern, L3Cube - Research Lab

Jul 2023 – Jun 2024

- Spearheaded the fine-tuning and optimization of a low-resource foundation model (Gemma-2B) using HuggingFace and PEFT (LoRA), leveraging GPU-accelerated training pipelines for efficiency and scalability.
- Realized a 65% accuracy gain on AI4Bharat benchmarks through manual QA evaluations that challenged conventional evaluation metrics for low-resource and code-switched LLMs, with outcomes published in peer-reviewed venues (IEEE, ACL, Springer).
- Architected and evaluated bilingual word embeddings with Hing-BERT, Hing-FastText, and Hing-GPT, delivering a 20% performance boost over standard baselines in social media hate speech detection and highlighting the advantages of domain-specific multilingual representations.

Data Science Intern, FeyNN Labs

Mar 2023 – Jun 2023

- Built an EV market analysis pipeline on consumer behavior, charging, and adoption trends; used clustering, forecasting, and feature engineering to improve accuracy by 5.5% and guide EV adoption planning.
- Authored a research report on AI-driven Predictive Maintenance systems in manufacturing, exploring ML methods (classification, anomaly detection, RUL estimation) for minimizing downtime and improving equipment reliability.

EDUCATION

Stony Brook University

Master of Science in Data Science, GPA: 3.70/4.00

08/2024 – 05/2026

Long Island, NY

Pune Institute of Computer Technology

Bachelor of Engineering, GPA: 9.04/10.00

08/2020 - 06/2024

Pune, India

PROJECTS

Deep Learning for Order Book Prediction | Quantitative Finance, BiGRU

May 2025

- Executed a High-Frequency Trading (HFT) workflow with performance tuning on real-time Binance BTC-USDT order book data, engineering microstructure features and training BiGRU sequence algorithms to estimate buy pressure (F1 87%) and estimate immediate trading costs (spread, slippage) for optimization.

Terrorism Risk Forecasting | R, Statistical Analysis, ML

Apr 2025

- Innovated an analytical process on global terrorism data applying time-series forecasting, bootstrap resampling, and SMOTE-enhanced classification, uncovering key patterns in attack frequency and weapon-target lethality.

Multilingual Named Entity Recognition | BiLSTM-CRF, HingBERT, Attention-based

Feb 2024

- Refined, validated a BiLSTM-CRF predictor with HingBERT encodings for Hinglish NER, attaining 85% F1-score on social media text; benchmarked against baseline BiLSTM and attention-based models, demonstrating the value of sequence processing and underrepresented, fused-dialect NLP.

PUBLICATIONS

Code-Mixed Embeddings for Hate Speech Detection – Springer ICCCN 2024

Jun. 2025

Adapting LLMs to Low-Resource Settings – ACL CHIpsAL 2025

Jan. 2025

Code-Mixing in Named Entity Recognition – IEEE ICETITE 2024

Feb. 2024

TECHNICAL SKILLS

Programming/Data: Python, R, C/C++, SQL, Java, React, Node.js, Pandas, NumPy, Folium.

ML/Platforms: NLP, MCP tool orchestration, RAG, LLMs, LangChain, Vector Databases, Pandas, NumPy, Sentence-Transformers, HuggingFace, Prompt Engineering, PEFT, MLOps, SQLite, Kubernetes, Docker, CI/CD, FastAPI